

Interactive Exploration of Annual Mean NO₂ across Europe

A Europe-Wide Overview with a Germany-Focused View

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Abstract — This report presents an interactive, coordinated-view visualisation system for exploring how nitrogen dioxide (NO₂) varies across space, time, and monitoring-station context across Europe. As dataset, the system uses the European Environmental Agency (EEA) Air Quality In-Situ Measurement Station Data published on Zenodo. The focus was on annual mean NO₂ values (2015–2023) together with station details (country, station type, station area) and geographic coordinates. The interface provides two geographic scopes—a Europe-wide overview and a Germany-specific view—linked by a global year slider to support year-by-year exploration. The system consists of coordinated views, which enable users to (i) identify spatial hotspots on maps, (ii) compare station-type distributions for the selected year via boxplots, (iii) assess long-term changes with trend lines aggregated by station type, and (iv) rank the top polluted countries for the chosen year. Limitations include the absence of cross-view brushing/linking and the sensitivity of min–max boxplot whiskers to outliers.

1 Introduction

Nitrogen dioxide (NO₂) is a major air pollutant and is commonly associated with combustion processes such as road traffic and industrial activities. Understanding how NO₂ varies across *space* (where are the hotspots?), *time* (do levels improve?), and *station context* (how the levels are different for traffic vs. background vs. industrial stations?) is important for air quality monitoring and discussions.

In this project, I build an interactive visualization for the *EEA Air Quality In-Situ Measurement Station Data* published on Zenodo [1]. My visualisation system focuses on annual mean NO₂ values and provides coordinated views at two geographic scopes: a Europe-wide overview (“World”) and a Germany-focused view. A year slider enables year-by-year exploration, while distribution, trend, and ranking views support comparison tasks.

Figure 1 shows both the Europe-wide (“World”) and Germany-focused views for an example year. It comprises a station map, a distribution summary by station

type, and long-term trends. Additional views, including a Europe-wide dashboard and a country ranking of the top polluted countries for the selected year, are shown in Fig. 2. Table 1 summarizes the key data attributes used throughout the system.

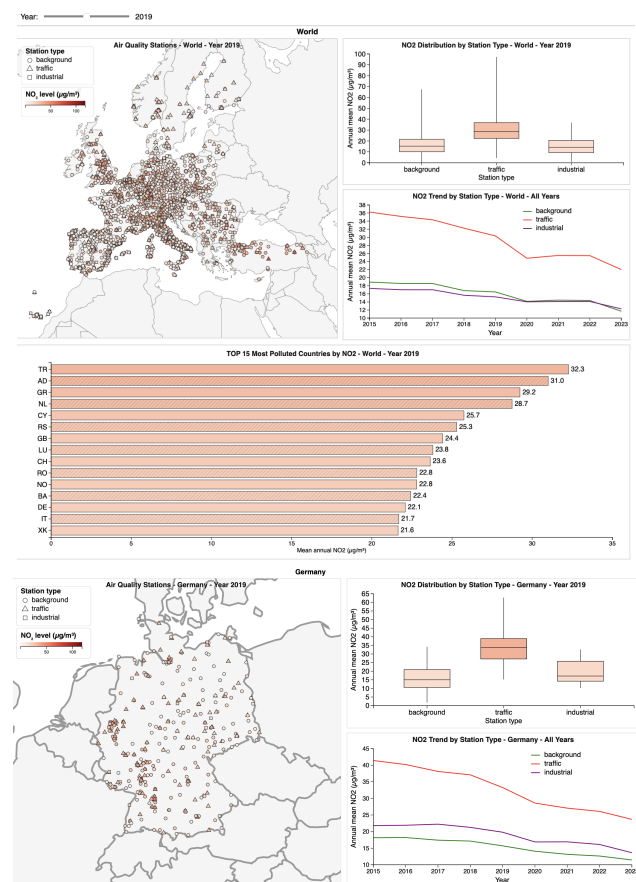


Figure 1 (top) Europe-wide (“World”) overview; (bottom) Germany-focused view. Both show annual mean NO₂ map (left panel), distribution by station type (top right), and trends (bottom right) for an example year.

2 Data abstraction

The primary items are station-year records from EEA monitoring stations across Europe. Each record includes NO₂ concentration (quantitative, µg/m³, range ≈ 0–100+), year (temporal, e.g., 2015–2023), station type (categorical: background/traffic/industrial),

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country code (categorical), and coordinates (spatial: longitude/latitude). Processed data are loaded from a CSV (converted from Parquet) and used in two ways: (1) year-specific subsets (filtered by year) are used for the maps, boxplots, and the country ranking; and (2) aggregated statistics across all years (computed via grouping and mean aggregation) are used for the trend lines by station type. Other pollutants (e.g., PM2.5, O₃) are not used in the core views to avoid issues with missingness and to keep the design focused on a single, consistently available measure.

3 Task abstraction

The design supports four domain-relevant tasks for environmental analysis:

(T1) Compare NO₂ distributions by station types in a given year. User can compare the NO₂ distributions for each station type; for example, it is possible to answer questions such as "Do traffic stations have higher medians than background stations in 2020?" This task is supported by the distribution view (boxplots by station type).

(T2) Locate NO₂ hotspots by station type. Users identify regions with high NO₂, for example "Where are high-traffic pollution clusters?" This task is supported by the station map view with color encoding by NO₂ level and shape encoding by station type.

(T3) Observe NO₂ trends over time by station type. Users check whether pollution has improved or worsened over the period, e.g., "Did industrial stations improve after 2018?" This task is supported by the trend line chart aggregating mean NO₂ by station type over all years.

(T4) Rank and compare top polluting countries in a selected year. Users ask questions like "Which countries had the highest annual NO₂ in 2019?" This task is supported by the top-polluted-countries bar chart presented for the selected year.

4 Technical description

This section describes the implemented views, their visual encodings, and the supported interactions.

4.1 Views

The dashboard has four views (Fig. 1, Fig. 2):

(V1) Station map (Europe-wide and Germany). Stations are shown as point symbols at their geographic coordinates on a base map. There are a Europe-wide

map and a Germany-focused map, which enable users to inspect spatial patterns at different scales.

(V2) Distribution by station type (year-specific). A boxplot summarizes the distribution of the annual mean NO₂ for each station type (background, traffic, industrial) in the currently selected year. This view supports side-by-side comparisons of main patterns and the difference between station types.

(V3) Trend by station type (all years). A multi-line chart shows the mean annual NO₂ over time for each station type. Unlike the map and boxplot, the trend view intentionally shows the full temporal context across all years.

(V4) Top polluted countries (year-specific). A horizontal bar chart shows the top 15 countries by mean annual NO₂ in the selected year. This enables a comparison across countries.

4.2 Visual encoding

The design uses the following visual encodings:

Maps (V1). Geographic *position* encodes station location (longitude/latitude) using a Mercator projection. NO₂ magnitude is encoded with a *sequential* color scale (light-to-dark red). Station type is encoded with *shape* (circle=background, triangle=traffic, square=industrial) to easily identify the category.

Boxplots (V2). Station type is encoded by *x-position* (grouped categories), while NO₂ is encoded by *y-position*. For each group, the box shows the interquartile range and the median line; whiskers extend to the minimum and maximum values of that group.

Trend lines (V3). Year is encoded by *x-position* and mean annual NO₂ by *y-position*. Station type is encoded by *color* (green=background, red=traffic, purple=industrial).

Country ranking (V4). Countries are encoded by *y-position* (rank order), while mean annual NO₂ is encoded by *bar length*. A light hatch overlay is used as a secondary styling cue to improve row separation.

4.3 Interactivity

The system supports the following interactions:

Year slider (global). A single year slider updates all year-dependent views: station maps, boxplots, and the country ranking. The current year value is displayed next to the slider to keep state visible.

Map pan/zoom. Both maps support pan and zoom. To preserve readability across zoom levels, station symbols are re-scaled such that their apparent size remains stable while zooming.

Attribute	Type	Example/Domain
NO ₂	Quantitative	0–100 $\mu\text{g}/\text{m}^3$
year	Temporal	2015–2023
Station.Type	Categorical	Background, Traffic, Industrial
Countrycode	Categorical	DE, IT, FR
Longitude/Latitude	Spatial	Geographic Coordinates (mostly inside Europe)

Table 1 Key attributes used in the visualization.

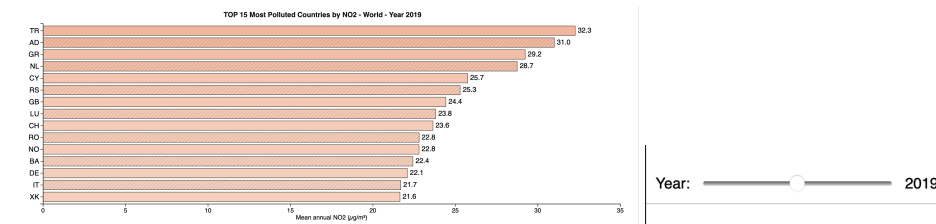


Figure 2 Top-15 countries by mean annual NO₂ [left]; year slider (global coordination across all views) [right].

Details-on-demand (tooltips). Hovering over a station symbol reveals a tooltip, which shows station identifier, country, station type, and the NO₂ value for the selected year.

Hover highlight (country bars). Hovering over a country bar increases its stroke width, providing a very simple focus+context mechanism for better user experience.

5 Evaluation

I evaluate the design with respect to the intended tasks (Section 3) and visualisation principles (effective channels, clarity, and readable comparisons).

Suitability. First, the station maps (V1) use geographic position—a highly effective channel in this context—to support hotspot identification (T2) (Fig. 2).

Second, the boxplot view (V2) supports comparisons between station types (T1) using a shared axis (Fig. 1). Third, the trend view (V3) provides temporal context across all years, which helps users judge whether the year-specific differences occur within a broader upward or downward pattern (T3).

State and consistency. The year slider creates a clear and consistent interaction pattern across the year-dependent views. Legends explicitly document the meaning of shape (station type) and color (NO₂ level), which makes the visualisation clear and easy to use. Consistent station-type categories (background/traffic/industrial) are reused across map, box-

plot, and trend views, which makes the whole system incredibly easy to learn.

Limitations. The current implementation does not include cross-view selection or brushing; thus, users cannot select a subset of stations on the map and see it highlighted in the other views. In addition, the trend view shows aggregated means across all years and does not change with the year slider; this choice reduces direct coordination between trend and year-specific plots. Finally, min-max whiskers in the boxplot can be sensitive to outliers; future work could use more robust boxplot type and/or show outlier points explicitly.

5.1 Case Study

An environmental analyst wants to understand the air quality across Europe. He/she begins by selecting a year of interest with the year slider. With the station map, the analyst can find areas with elevated NO₂. In addition, the country ranking (Fig. 2) helps the analyst quickly identify which countries are among the highest in that year (T4).

Next, the analyst switches to the Germany view (Fig. 1, bottom) to inspect a single country at higher spatial resolution (T2). Hovering over stations reveals details-on-demand (station id, type, and NO₂ value), which helps validate whether high values are associated with particular station types or local pollution clusters. The boxplot then supports a direct comparison between station types for that same year (T1), while the trend lines provide longer-term context across all years to understand whether the observed

year represents part of a steady decline, increase, or a more stable pattern (T3).

6 Conclusion

5 This report presented an interactive visualisation for exploring annual mean NO₂ measurements from EEA monitoring stations [1]. The visualisation combines spatial context (maps), category comparison (box-plots), temporal context (trend lines), and macro-level comparison (top-country ranking). Users can explore
10 how NO₂ varies by location, station type, and time. Future work could add cross-view selection (brushing/linking) and better ways to display data distributions and outliers.

References

- 15 [1] European Environment Agency and Johannes Heisig. Eea air quality in-situ measurement station data, 2024.